Project: Worked examples
Title: Shaft under combined bending and torsion
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the bending and torsional stresses in a solid round shaft, with stress-concentration factors, and the factors of safety against yield. It covers static loading only, not fatigue or hollow shafts.

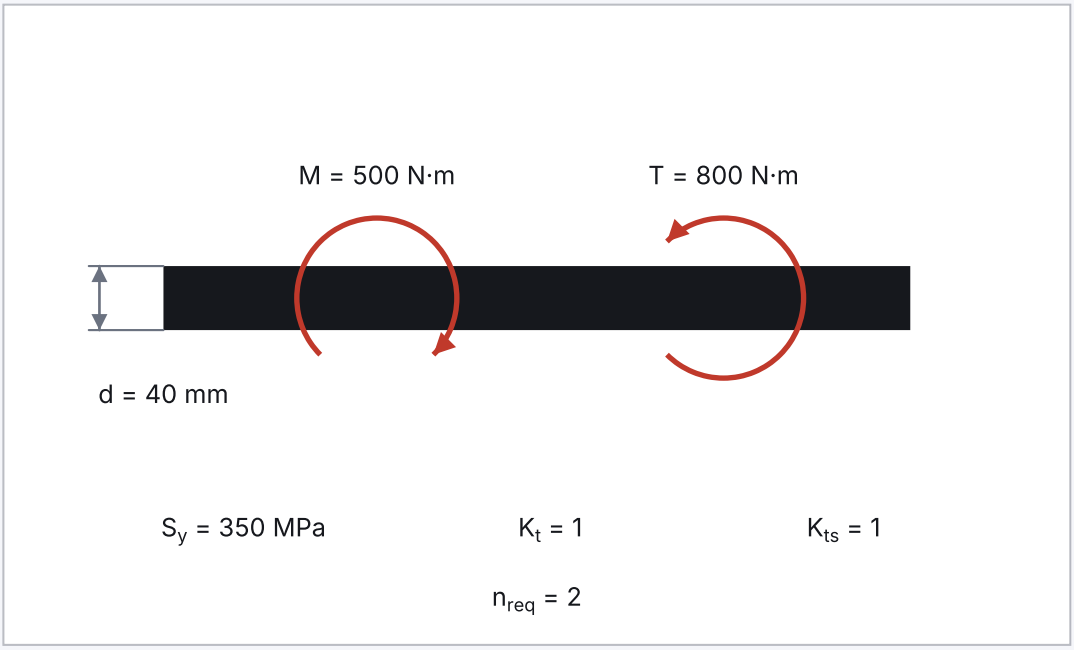


Figure 1 — A solid round shaft under bending and torsion at the section. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Bending stress, with its stress concentration	$\sigma = K_t \cdot 32M/(\pi d^3)$
Torsional shear stress, with its stress concentration	$\tau = K_{ts} \cdot 16T/(\pi d^3)$
Maximum shear stress (Tresca)	$\tau_{max} = \sqrt{((\sigma/2)^2 + \tau^2)}$
von Mises equivalent stress	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$
Factors of safety against yield, the lower governing	$n_T = (S_y/2)/\tau_{max}$; $n_{VM} = S_y/\sigma'$

Basis: Maximum-shear-stress and distortion-energy theories[1](§5-4 and §5-5, 10th ed.); Shaft design for stress[1](§7-4, 10th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Shaft diameter	40	mm	drawing: DRG-SH-401 rev A	1 … 2000 mm
S_y	Yield strength	350	MPa	drawing: DRG-SH-401 rev A	50 … 3000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
M	Bending moment (magnitude)	500	N·m	spec: LOAD-SCHEDULE-01	0 … 10000000 N·m
T	Torque (magnitude)	800	N·m	spec: LOAD-SCHEDULE-01	0 … 10000000 N·m
K_t	Bending stress-concentration factor (1 for none)	1		drawing: DRG-SH-401 rev A	1 … 5
K_{ts}	Torsional stress-concentration factor (1 for none)	1		drawing: DRG-SH-401 rev A	1 … 5
n_{req}	Required factor of safety	2		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. The shaft is a solid round section, linear elastic, under static loading. (The stress formulas of the method assume this.)

4. Working

[8] $\sigma = K_t * 32 * M / (\pi * d^3)$ [1](§7-4)

$= 1 \times 32 \times 500\ \text{N·m} / (\pi \times (40\ \text{mm})^3)$

$= 79.577\ \text{MPa}$

Bending stress

[9] $\tau = K_{ts} * 16 * T / (\pi * d^3)$ [1](§7-4)

$= 1 \times 16 \times 800\ \text{N·m} / (\pi \times (40\ \text{mm})^3)$

$= 63.662\ \text{MPa}$

Torsional shear stress

[10] $\tau_{max} = \text{sqrt}((\sigma / 2)^2 + \tau^2)$ [1](§5-4)

$= \text{sqrt}((79.577\ \text{MPa} / 2)^2 + (63.662\ \text{MPa})^2)$

$= 75.073\ \text{MPa}$

Maximum shear stress (Tresca)

[11] $\sigma_{vm} = \text{sqrt}(\sigma^2 + 3 * \tau^2)$ [1](§5-5)

$= \text{sqrt}((79.577\ \text{MPa})^2 + 3 \times (63.662\ \text{MPa})^2)$

$= 135.98\ \text{MPa}$

von Mises stress

[12] $n_T = (S_y / 2) / \tau_{max}$

$= 350\ \text{MPa} / 2 / 75.073\ \text{MPa}$

$= 2.3311$

Factor of safety, maximum shear stress

[13] $n_{VM} = S_y / \sigma_{vm}$

$= 350\ \text{MPa} / 135.98\ \text{MPa}$

$= 2.5739$

Factor of safety, distortion energy

[14] $n_{min} = \min(n_T, n_{VM})$

$= \min(2.3311, 2.5739)$

$= 2.3311$

Governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
τ_{max}	75.07 MPa	Maximum shear stress
σ_{vm}	136.0 MPa	von Mises stress
n_{min}	2.331	Governing factor of safety
σ	79.58 MPa	Bending stress
τ	63.66 MPa	Torsional shear
n_T	2.331	Factor of safety (Tresca)
n_{VM}	2.5738	Factor of safety (von Mises)
u_n	0.86	Utilisation of factor of safety against the required factor (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Factor of safety against the required factor	2	2.3311	0.86	[1](§7-4)

7. Applicability

None.

8. Verification

Template mech.shaft-stress@1.1.1 Content hash e68a699ed74d32f0 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.